AI FOR SOCIAL IMPACT / MARKET DESIGN

Who keeps an offer?

A better prediction can leave more applicants without a place.

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The five-choice test

Families can see five programs plus any guaranteed current-school option. Does keeping the program a family was actually offered on its list help more pupils get a place?

One model input changed

Two logistic models learned from 2019 offers. Both saw the programs a family listed. One also saw their order. We froze both models and specified the expected result before opening either test cohort.

We counted how often the five-choice rule retained a recorded offer, then replayed admissions. In deferred acceptance, each pupil applies down their visible list and each program keeps its highest-priority applicants up to capacity.

Both learned rules beat uniform lists in the earlier cohorts. The comparison here isolates the addition of rank information.

Two new cohorts gave the same answer

Changes from adding rank to the predictor, across 1,107,244 applicant records in Chile.

+2,493

historical offers shown

−985

placements in the replay

Cohort	Additional offers shown	Change in placements
2022	**+1,143**	**−468**
2023	**+1,350**	**−517**

Of the net increase in covered offers, 2,244 were offers families explicitly accepted. In the replay, 1,455 pupils lost a place and 470 gained one.

Table from the Chilean Ministry of Education's school admissions records [1].

Which vacancies can a student use?

In a separate replay for 189,399 university applicants, 21,061 seats were empty. Of 30,866 unmatched applicants, just 160 had listed an eligible program with a vacancy.

An exact limit on safe referrals

Connect each unmatched applicant to the vacancies on their original list. A maximum matching assigns the most referrals that fit while keeping every tentative contract feasible.

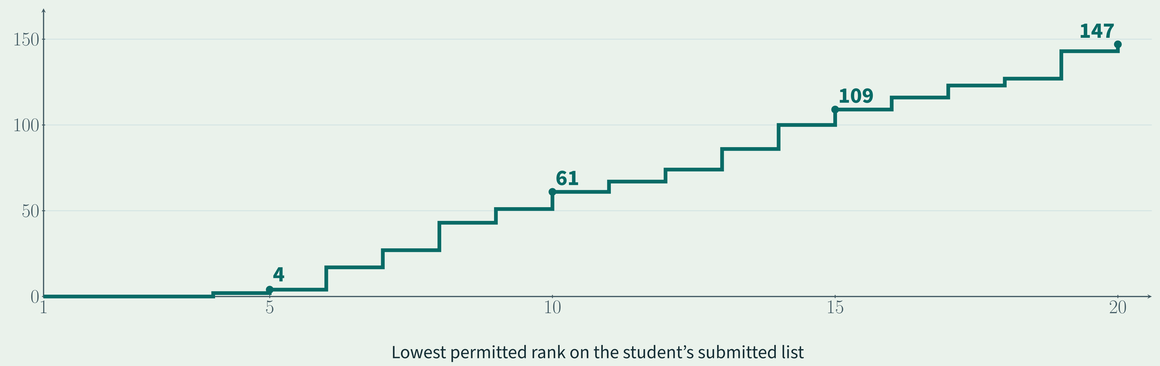
160 − (24 − 11) = 147

A group of 24 applicants shared just 11 reachable seats. One referral per selected applicant attained the bound. Everyone already placed kept a program they ranked at least as highly.

How far down the list?

If a referral had to be in the student's first five choices, only four people could gain a place. At a rank limit of ten, the maximum rose to 61.

Maximum additional placements



Graph from the project's exact matching calculations on 2026 regular admissions [2].

Observed data include submitted lists, offers and responses. Placements under changed recommendations are counterfactual.

[1] Chile MINEDUC, Sistema de Admisión Escolar, 2022–23 applications. [2] Chile MINEDUC, PAES 2026, regular applications and published program seats.

Harvard CS 2880, AI for Social Impact. Poster, table and graph created with OpenAI Codex and LaTeX, 13 November 2026.